DP024
Physics 2
Semester 2
Session 2024/2025
2 hours

DAQ SET A

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PROGRAM DRAW A QUESTION
MODEL PHYSICS QUESTION PAPER
2 HOURS

DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

INSTRUCTIONS TO CANDIDATE:

The question paper consists of **6** question.

Answer **all** questions.

The use of non-programmable scientific calculator is permitted.

QUESTION	MARKS
1	
2	
3	
4	
5	
6	
TOTAL	

*This question paper consists of **10** pointed pages.*

LIST OF SELECTED CONSTANT VALUES

SENARAI NILAI PEMALAR TERPILIH

Speed of light in vacuum *Laju cahaya dalam vakum*	c	$= 3.00\times10^{8}\ m\ s^{-1}$
Permeability of free space *Ketelapan ruang bebas*	μ_o	$= 4\pi\times10^{-7}\ H\ m^{-1}$
Permittivity of free space *Ketelusan ruang bebas*	ε_o	$= 8.85\times10^{-12}\ F\ m^{-1}$
Electron charge magnitude *Magnitud cas elektron*	e	$= 1.60\times10^{-19}\ C$
Planck constant *Pemalar Planck*	h	$= 6.63\times10^{-34}\ J\ s$
Electron mass *Jisim elektron*	m_e	$= 9.11\times10^{-31}\ kg$ $= 5.49\times10^{-4}\ u$
Neutron mass *Jisim neutron*	m_n	$= 1.674\times10^{-27}\ kg$ $= 1.008665\ u$
Proton mass *Jisim proton*	m_p	$= 1.672\times10^{-27}\ kg$ $= 1.007277\ u$
Deuteron mass *Jisim deuteron*	m_d	$= 3.34\times10^{-27}\ kg$ $= 2.014102\ u$
Molar gas constant *Pemalar gas molar*	R	$= 8.31\ J\ K^{-1}\ mol^{-1}$
Rydberg constant *Pemalar Rydberg*	R_H	$= 1.097\times10^{7}\ m^{-1}$
Avogadro constant *Pemalar Avogadro*	N_A	$= 6.02\times10^{23}\ mol^{-1}$
Boltzmann constant *Pemalar Boltzmann*	k	$= 1.38\times10^{-23}\ J\ K^{-1}$

Gravitational constant *Pemalar graviti*	G	$= 6.67 \times 10^{-11}\ N\,m^{2}\,kg^{-2}$
Free-fall acceleration *Pecutan jatuh bebas*	g	$= 9.81\ m\,s^{-2}$
Atomic mass unit *Unit jisim atom*	1 u	$= 1.66\times10^{-27}\ kg$ $= 931.5\ \frac{M\,eV}{c^{2}}$
Electron volt *Elektron volt*	1 eV	$= 1.6\times10^{-19}\ J$
Constant of proportionality *Pemalar hukum Coulomb*	$k = \frac{1}{4\pi\varepsilon_o}$	$= 9.0 \times 10^{9}\ N\,m^{2}\,C^{-2}$
Atmospheric pressure *Tekanan atmosfera*	1 atm	$= 1.013 \times 10^{5}\ Pa$
Density of water *Ketumpatan air*	ρ_w	$= 1000\ kg\,m^{-3}$

LIST OF SELECTED FORMULAE
SENARAI RUMUS TERPILIH

1. $a =- \omega^2$

2. $y = A \sin sin\ \omega t$

3. $\omega = \frac{2\pi}{T} = 2\pi f$

4. $v = A\omega \cos cos\ \omega t$

5. $a =- A\omega^2 \sin sin\ \omega t$

6. $k = \frac{2\pi}{\lambda}$

7. $y(x, t) = A \sin sin\ (\omega t \pm kx)$

8. $v = \frac{dy}{dt}$

9. $v = f\lambda$

10. $y = 2A \cos cos\ kx\ \sin sin\ \omega t$

11. $v = \sqrt{\frac{T}{\mu}}$

17. $\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + ... + \frac{1}{R_n}$

18. $V = IR$

19. $C = \frac{Q}{V}$

20. $\frac{1}{C_{eff}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + ... + \frac{1}{C_n}$

21. $C_{eff} = C_1 + C_2 + C_3 + ... + C_n$

22. $\tau = RC$

23. $Q = Q_o e^{-\frac{t}{RC}}$

24. $Q = Q_o\left(1 - e^{-\frac{t}{RC}}\right)$

25. $B = \frac{\mu_o I}{2\pi r}$

26. $B = \frac{\mu_o I}{2r}$

27. $B = \mu_o nI$

12. $\mu = \frac{m}{l}$

13. $I = \frac{dQ}{dt}$

14. $Q = ne$

15. $\rho = \frac{RA}{l}$

16. $R_{eff} = R_1 + R_2 + R_3 + ... + R_n$

28. $\phi = BA \cos cos\, \theta$

29. $\Phi = N\phi$

30. $\varepsilon = - \frac{d\phi}{dt}$

31. $\varepsilon = Blv \sin sin\, \theta$

32. $\varepsilon = - NA\frac{dB}{dt}$

33 $\varepsilon = - NB\frac{dA}{dt}$

34. $R = 2f$

35. $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

36. $m = \frac{h_i}{h_o} = - \frac{v}{u}$

37 $y_m = \frac{m\lambda D}{d}$

38. $y_m = \frac{\left(m + \frac{1}{2}\right)\lambda D}{d}$

39. $\Delta y = \frac{\lambda D}{d}$

Answer all the questions.

1. (a)

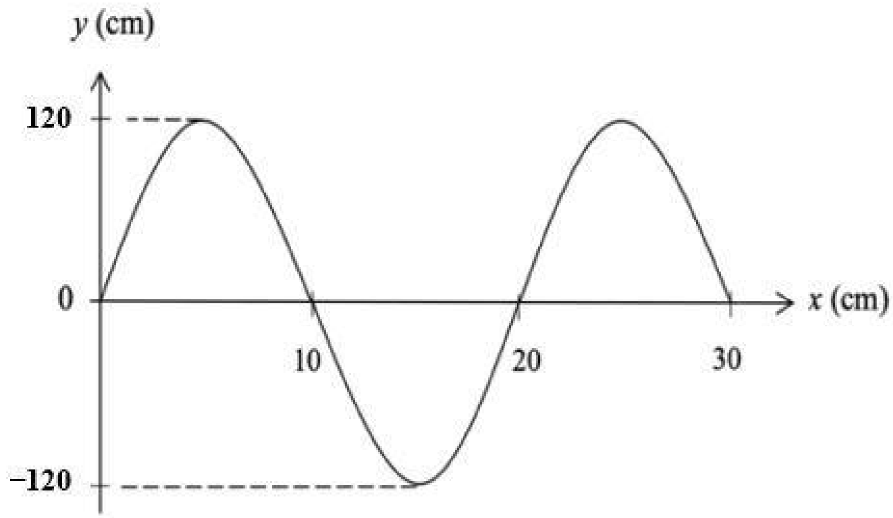

FIGURE 1

FIGURE 1 above shows a displacement- distance graph with a speed of wave is 3.00 m s^{-1} moving to the right.

(i) Determine the maximum displacement and the wavelength.

(ii) Write the wave equation.

[7 *marks*]

(b) A standing wave is represented by the equation below where Y and x are in metre and t in milliseconds.

$$Y = 6.0 \sin sin\, 20t \cos cos\, 40x$$

(i) Write two prograssive waves expression .

(ii) Calculate the particles's amplitude at x = 2 m of the stationary wave.

[6 *marks*]

(c) A stretched string is vibrates with length of 50 cm, mass 1.0 g and tension 100 N. Calculate the speed of the transverse waves travelling along the string.

[3 *marks*]

2.

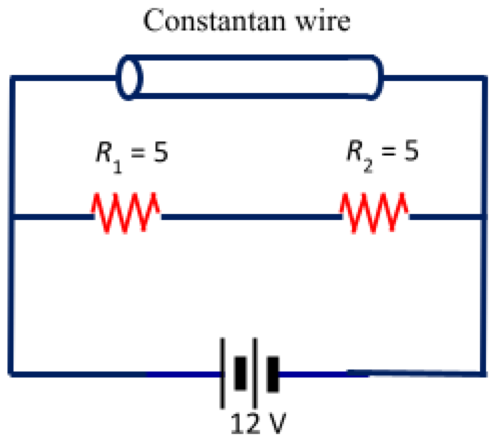

FIGURE 2

FIGURE 2 shows a constantan wire of length 1.6 m and cross-sectional wire area of 0.12 mm^2 that is connected to two identical resistors with a resistance 5 Ω and a 12 V power supply. The resistivity of constantan wire at room temperature is 4.0×10^{-7} Ω m. Calculate the

a) resistance of the constantan wire.

b) current flowing through the constantan wire.

c) current and total charge flowing through resistor R_1 in 2 s.

[12 *marks*]

3. (a)

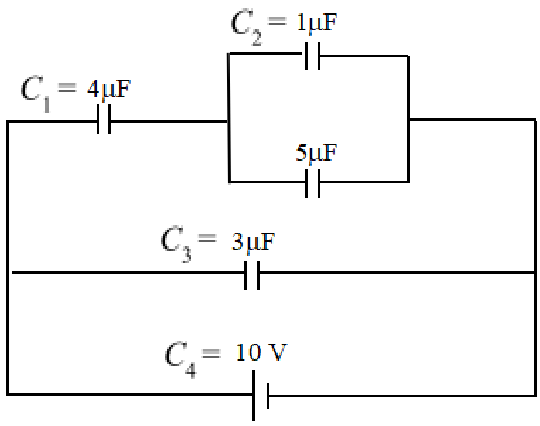

FIGURE 3a

Four capacitors are connected to a battery as shown in **FIGURE 3a**. Calculate the

(i) Equivalent capacitance of the arrangement of capacitors in the circuit.

(ii) Charge on the capacitor of 4 μF.

(iii) Potential difference of 5μF.

[5 *marks*]

(b)

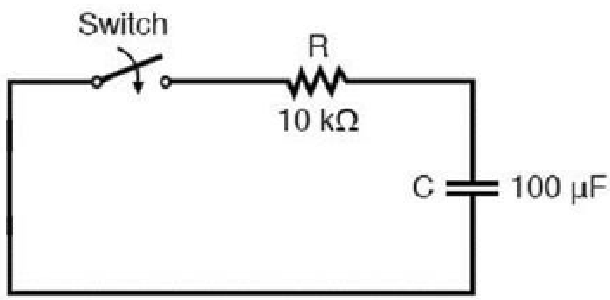

FIGURE 3b

FIGURE 3b shows a circuit of capacitor and resistor. The capacitor is fully charged. When the switch is closed, the capacitor is discharging. Calculate the

(i) Time taken for the charge reduced to 45% of its maximum value.

(ii) Sketch the graph of current I against time, t for this process.

[5 *marks*]

4. (a)

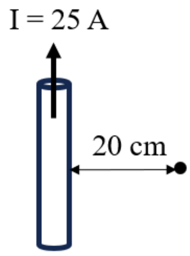

FIGURE 4

FIGURE 4 shows a long straight wire placed vertically carries an upward current of 25 A. What is the magnitude and direction of the magnetic field strength at a point 20 cm to the right of the wire?

[3 *marks*]

(b) A coil of surface area 20 cm^2 is placed in a uniform magnetic field of 5 T. Find the magnetic flux that crosses the coil when the plane of the coil

(i) Is perpendicular to the magnetic field.

(ii) Is making an angle of 30° to the magnetic field.

[5 *marks*]

(c) A circular coil of 25 turns and radius 20 cm is placed in an external magnetic field of strength 0.2 T so that the plane of the coil is perpendicular to the field. Find the emf induced in the coil when

(i) The coil is pulled out of the field in 0.25 s.

(ii) The radius of the coil decreases to 10 cm in 0.2 s.

[6 *marks*]

5. (a) A convex mirror with a focal length of 15.0 cm is placed 35.0 cm from a lighted object.

(i) Calculate the image distance.

(ii) Calculate the linear magnification

(iii) Sketch a labelled ray diagram for the formation of image and state **TWO (2)** characteristics of image formed.

[9 *marks*]

(b) An object with height 15.0 mm is placed in front of the thin lens at a distance of 10.0 cm. A real image formed is located 20.0 cm away from the lens.

(i) Calculate the focal length of the and state the type of lens is used.

(ii) Calculate the height of image formed.

[5 *marks*]

6. (a) In Young's double-slit experiment, the light source emits the blue light of wavelength $420\ nm$. The slits are $0.08\ mm$ apart and interference pattern is observed on a screen at a distance of 65 cm from the slits.

Calculate the

(i) separation between two consecutive dark fringes.

(ii) distance between the central maximum and sixth dark fringes.

(iii) new wavelength of the light source if ten maximum orders of bright fringes with separation as in b (ii) formed on the screen.

[8 *marks*]

(b) A third order of constructive interference for double slit pattern with distance 18 cm from central maximum is viewed on a screen 1.2 m from the slits.

(i) Calculate the ratio of wavelength to separation between slits

(ii) How would you observe the interference pattern if the entire experiment is submerged in water?

[6 *marks*]

END OF QUESTION PAPER

KERTAS SOALAN TAMAT